Supplementary materials

Dataset 1

Raw data collected during field surveys by observers and from automatic recorders at two sites. The dataset includes:

- Recorder – recorder ID number;
- Date – date of the survey;
- Hour – time of the field survey conducted by the observer;
- Pond ID – name of the waterbody;
- N_males – number of males detected by the observer during the field survey;
- ACI_sec_hour – average value of the *Acoustic Complexity Index* (*ACI*) for the second hour after sunset;
- BI_sec_hour – average value of the *Bioacoustic Index* (*BI*) for the second hour after sunset;
- ACI_night – average *ACI* value for the entire night (from sunset to sunrise);
- BI_night – average *BI* value for the entire night (from sunset to sunrise);
- N_call_sec_hour – number of calls automatically detected during the second hour after sunset;
- N_call_night – number of calls detected throughout the entire night (from sunset to sunrise).

Dataset 2

Raw data from automatic recorders at three sites. The dataset includes:

- Recording – the name of the recording;
- Date – the date of the recording;
- Time – the start time of the recording;
- ACI – the average *Acoustic Complexity Index* (*ACI*) value for the one-hour recording;
- BI – the average *Bioacoustic Index* (*BI*) value for the one-hour recording;
- Temperature – the average temperature (in degrees Celsius) for the hour (values recorded by the automatic sound recorder);
- N_call – the number of vocalizations automatically detected during the hour.

Call.wav

An example of a European green toad call. The call was used to generate the spectrogram presented in Fig. 1.